A. Flip Flop Sum

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: greedy, implementation | https://codeforces.com/problemset/problem/1778/A

You are given an array of n integers $a_1, a_2, ..., a_n$. The integers are either 1 or -1. You have to perform the following operation **exactly once** on the array a:

- Choose an index i ($1 \le i$ lt n) and flip the signs of a_i and a_{i+1}. Here, flipping the sign means -1 will be 1 and 1 will be -1.

What is the maximum possible value of $a_1 + a_2 + ... + a_n$ after applying the above operation?

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 500$). Description of the test cases follows.

The first line of each test case contains a single integer n ($2 \le n \le 10^5$), the length of the array a.

The next line contains n integers $a_1, a_2, ..., a_n$ ($a_i = 1$ or $a_i = -1$).

The sum of n over all cases doesn't exceed 10^5.

Output

For each test case, print the maximum possible sum of the array a you can get in a separate line.

Example

Input
4 5 -1 1 1 -1 -1 5 1 1 -1 -1 -1 2 1 1 4 1 -1 -1 1

Output
3 3 -2 4

Note

In the first case, we can choose index 4 and flip the signs of a_4 and a_5. After this operation, the sum will be -1+1+1+1+1 = 3. We can't make the sum larger than this.

In the third case, the only option is to choose the index 1.